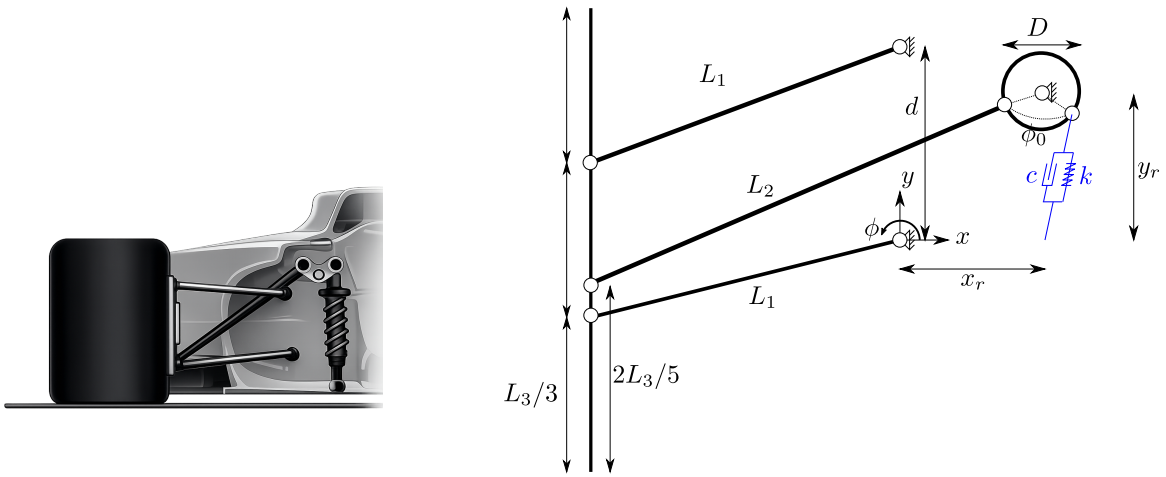
MECA0031-2
Kinematics and dynamics of mechanisms
Academic year 2025-2026

First practical work: F1 pushrod suspension

1 Description of the problem



(a) (b)

Figure 1: Schematic view of a pushrod suspension. Scales not respected.

A pushrod suspension is a suspension device commonly used in Formula 1. As all suspensions, its prime role is to avoid the car body to feel any perturbation on the road by always keeping it horizontal with respect to the ground and by damping any oscillations. A schematic view of this kind of suspension is presented in Figure 1. The main goal of this project is to analyse the dynamic response of a F1 pushrod suspension device to an applied force on one wheel in order to simulate a bump. To do so, it is asked to study a simplified planar model with the following assumptions:

1. the lower and upper wishbone (both of length L_1), pushrod (L_2), and wheel (L_3) are modelled as rigid bars. The centre of mass of each bar component i is located at its midpoint with rotational inertia $J_i = m_i l_i^2/12$;
2. elements are connected to each other with revolute joints for which friction is neglected;
3. the rocker is replaced by a rigid cylinder with rotational inertia $J_i = m_i D^2/8$;
4. the rocker is connected to a linear spring and damper element characterised by stiffness k, damping coefficient c and natural length l_0;
5. the wheel lower extremity experiences a vertical force: $F(t) = 10^2 \exp(-(t-0.5)^2)$ [N].

L_1	0.5	[m]	m_1	2.5	[kg]
L_2	0.6	[m]	m_2	2	[kg]
L_3	0.72	[m]	m_3	15	[kg]
D	0.15	[m]	m_4	1	[kg]
x_r	0.1306	[m]	y_r	0.1934	[m]
ϕ_0	120	[°]	d	0.25	[m]
k	$5 \cdot 10^4$	[N/m]	l_0	0	[m]

Table 1: Geometrical and physical parameters.

All relevant dimensions, masses, initial angles (defined with respect to the horizontal and positive anti-clockwise), and other necessary parameters are provided in Table 1.

This analysis will be performed by first establishing the equations of motion for the suspension model, and then writing a python/matlab program integrating in time the set of equations using Newmark scheme. Afterwards, using the numerical results, the choice of the optimal damping and stiffness of the linear spring will be discussed in order to limit and damp as quick as possible the amplitude of the oscillations of the car body when the F1 car encounters a bump at a given speed.

2 Questions

1. Calculate the number of degrees of freedom of the mechanism using the Grübler formula (max 1/2 page).

2. Establish the free-body diagram of the system. Define the finite element coordinates and the boundary conditions. Write down the assembly constraints and the rigid body constraints. Choose a set of active coordinates (max 4 pages, do not present numerical results here).

3. Write the equations of motion of the system and their linearized form (max 2 page).

4. Develop a python/Matlab program to solve the nonlinear equations of motion using the Newmark method (no description is required, only the well-commented python/Matlab code).

5. In the case of no stiffness and damping, what is the hard geometric limit of the suspension submitted to gravity? What do you observe, comparing to this configuration, when a spring with stiffness k and natural length $l_0 = 0$ is added? Note that one spring extremity is fixed at $(x_r, 0)$. At this new static equilibrium, apply a vertical force $F(t)$ on the lower end of the wheel and simulate the suspension for 2s. Show the evolution in time of the rocker angle, angular velocity and acceleration. Can you improve the suspension behaviour by adapting the characteristics of spring and damper? For this analysis, select adequate values for the numerical parameters and justify their choosing (max 3.5 pages).

3 Instructions

This work is to be performed in groups of two students. Each group must submit a brief report and a python/Matlab code. These documents should be sent by email to Nayan.Levaux@uliege.be

for March 31, 2026. The report should be written in a clear and concise manner, respecting the strict limit of 10 pages maximum (results and figures included but excluded front page and table of content). It is useless to recall the problem statement, make theoretical background and describe the python/Matlab code. Note that every student is susceptible to questioning on the project during the oral exam.